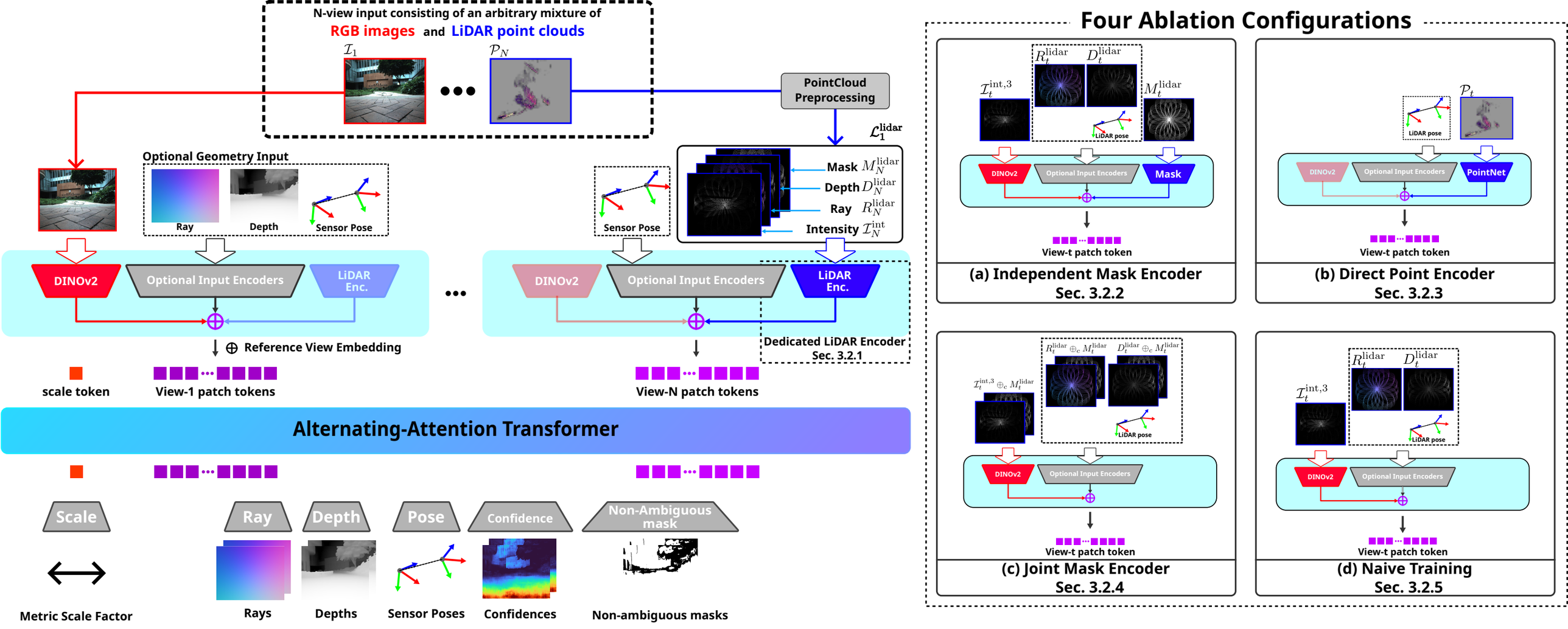

N-view input consisting of an arbitrary mixture of
RGB images and LiDAR point clouds
PointCloud Preprocessing
Optional Geometry Input
Ray
Depth
Sensor Pose
Mask
Depth
Ray
Intensity
DINOv2
Optional Input Encoders
LiDAR Enc.
Reference View Embedding
Dedicated LiDAR Encoder
Sec. 3.2.1
scale token
View-1 patch tokens
View-N patch tokens
Alternating-Attention Transformer
Scale
Ray
Depth
Pose
Confidence
Non-Ambiguous mask
Metric Scale Factor
Rays
Depths
Sensor Poses
Confidences
Non-ambiguous masks
Four Ablation Configurations
LiDAR pose
Mask
PointNet
View-t patch token
(a) Independent Mask Encoder
Sec. 3.2.2
(b) Direct Point Encoder
Sec. 3.2.3
(c) Joint Mask Encoder
Sec. 3.2.4
(d) Naive Training
Sec. 3.2.5